

Radiotherapy of Choroidal Malignant Melanomas

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Choroidal malignant melanomas are relatively rare tumours and can have a very long natural history. The management of these tumours is controversial. The results of a study of a small group of patients treated by radiotherapy, either as primary treatment or following local excision or enucleation, are presented here. The value of radical radiotherapy in the management of choroidal malignant melanoma is discussed.

Malignant melanoma of the eye is a relatively rare tumour. The incidence in the United Kingdom is 0.2 per 10⁵ population per year and in the United States of America 0.7 per 10⁵ population per year. Although relatively uncommon it is the commonest primary intra-ocular tumour. Eighty-five per cent arise in the choroid; the remainder arise in the ciliary body (9%) and in the iris (6%). The peak age incidence is in the 6th decade.

Tumour spread is mainly local. However, distant metastases occur in about 50% of patients. The time from the initial diagnosis to the detection of metastatic spread can vary from 1 month up to 20 years, with a mean of 4 years. This very variable natural history makes it difficult to outline a management policy for this tumour. Twenty per cent of the patients develop local recurrence following enucleation. The vast majority of these have extrascleral spread at the time of enucleation; local recurrence is rare in patients with no extrascleral spread. Patients who subsequently develop metastases die of their disease within 9 to 12 months.

Patients may present initially with visual symptoms associated with tumour growth or retinal detachment. Others may develop secondary glaucoma or pain in the orbit. In a number of cases the primary tumour causes no symptoms and is only detected when the metastases give rise to symptoms.

The results of a study of a small group of patients with malignant melanoma of the choroid referred for radiotherapy and treated at the Glasgow Institute of Radiotherapeutics and Oncology are presented here and the value of radiotherapy in relation to associated morbidity and the effect on survival is discussed.

MATERIALS AND METHODS

Twenty-eight patients with the diagnosis of malignant choroidal melanoma are presented here. There were 18 males and 10 females. The age range was from 31 to 83 years, the majority being in their 6th decade of life.

There were two groups of patients. The first group of 16 had small tumours (Table 1). Eight of these were treated with ⁶⁰Co plaques only, and eight were treated by local excision followed by application of a ⁶⁰Co plaque. In all these cases the ⁶⁰Co plaque was of a Stallard

type which was sewn on to the appropriate site on the sclera and left in place so that the equivalent dose at the tumour apex was approximately 80 Gy given over 14 days. In all but one case the diameter of the ⁶⁰Co plaque was 15 mm.

The second group comprised 12 patients who had large tumours. Five were treated by enucleation and post-operative external radiotherapy, seven were treated by enucleation only. In these 12 cases histology confirmed extrascleral spread. The dose given to the five patients receiving adjuvant post-operative radiotherapy to the orbit was between 55 Gy and 60 Gy in 30 fractions over 6 to 7 weeks using a linear accelerator.

RESULTS

Fourteen out of the 16 patients with small lesions were alive and well from 34 to 174 months after initial treatment (Table 1). One patient died of melanoma with liver metastases, and a second patient died of carcinoma of the bronchus. Pathological confirmation was not available in all the cases as only eight of the 16 had local excision carried out as part of the initial treatment. In the eight cases where there was no pathology initially, three eventually required enucleation. In this group of 16 patients four had tumours of spindle B type; six had mixed tumours and one had an epithelioid tumour.

Ten out of the 16 patients developed complications in the treated eye (62.5%). Two developed cataracts; three had vitreous haemorrhage; one developed both a cataract and a vitreous haemorrhage; others developed secondary glaucoma, blindness, vasculitis of the optic nerve and diminished visual activity. One patient developed such severe pain that the eye had to be enucleated. The high rate of complications is probably related to the size of the plaque, which was in the main 15 mm in diameter (Migdal, 1983).

The second group of patients were treated by enucleation with or without post-operative radiotherapy (Table 2). Only one of the patients who received prophylactic post-operative radiotherapy was alive at 63 months, the rest having died of distant metastases. In the group which did not receive adjuvant post-operative radiotherapy, only one developed a local recurrence (Table 2, Patient 6) 3 years after enucleation. This patient subsequently received radiotherapy to the orbit; he developed liver metastases and died 84 months after enucleation. Three of the other six patients are known to have died of distant metastases; one patient was lost to follow up and two patients were still alive with no evidence of disease 30 and 142 months after enucleation. In this group those with mixed tumours appeared to fare better than those with spindle cell B tumours. The survival for the small group was 57% (four out of seven) for